ModelIt!

Student Activity Pack

Can We Build a Fusion Reactor?

Modeling Plasma Confinement, Energy Balance, and the Path to Unlimited Clean Energy

Grade 11L3 | NGSS HS-PS1-8, HS-PS3-2 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. What is nuclear fusion and how does it differ from the nuclear fission used in today's power plants?

2. Why do you think it has taken over 70 years of research and billions of dollars without achieving commercial fusion power?

3. Draw a diagram of what you think happens inside a fusion reactor — what goes in, what comes out, and what keeps it working.

4. If fusion is the energy source of every star, what makes it so difficult to reproduce in a building on Earth?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Plasma Temperature | Magnetic Field Strength | Fuel Density | Energy Output | Neutron Flux | Wall Material Degradation | Net Energy Gain

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? **Explain ONE relationship in your model. What happens when one component changes?**

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: ITER Design Parameters

150 million degrees, 5.3 Tesla, moderate fuel density

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Push to Ignition

Maximum achievable temperature, field strength, and density

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Confinement Collapse

Magnetic field strength reduced by 25%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Tritium Breeding Ratio — The rate at which the lithium blanket surrounding the plasma produces new tritium fuel from neutron bombardment — the breeding ratio must exceed 1.0 or the reactor runs out of fuel, since natural tritium is virtually nonexistent

Plasma Disruption Frequency — The rate of sudden violent instabilities where confined plasma dumps its enormous thermal energy onto reactor walls in milliseconds — each disruption can severely damage first-wall components and requires repair downtime

Divertor Heat Load — The thermal power density on the exhaust component that handles plasma particles escaping confinement — divertor surfaces face heat loads comparable to spacecraft reentry, and no material can survive these conditions indefinitely

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why has the most abundant energy source in the universe — the same reaction that powers every star — remained just out of human reach for over 70 years? Use evidence from your simulation to support your answers.

1. **How does your model illustrate why the Lawson criterion makes fusion a fundamentally harder engineering challenge than fission?**

2. **Why does the energy-output-versus-wall-damage trade-off represent an inherent design tension rather than a solvable problem?**

3. **What does the confinement collapse scenario reveal about the stability margins of magnetically confined plasma?**

4. **How would your model predictions change if a material were discovered that was completely resistant to neutron damage?**

5. **What are the limitations of modeling fusion with seven components when a real reactor has thousands of interacting subsystems?**

8 STEM Challenge: Design a Next-Generation Fusion Reactor Approach

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YOUR ENGINEERING MISSION

Select the most critical engineering bottleneck preventing commercial fusion and design an innovative approach to overcome it, using evidence from your computational model to justify your strategy.

THE SCENARIO

The Department of Energy has launched a grand challenge for student teams to propose breakthrough approaches to commercial fusion energy. Your team must identify which variable in the fusion system represents the most critical bottleneck — plasma stability, neutron-resistant materials, tritium fuel supply, or energy balance optimization — and propose an engineering concept that could transform that variable, backed by quantitative evidence from your model.

Engineering Design Thinking

1. WHAT TYPE OF MAGNETIC CONFINEMENT DOES YOUR DESIGN USE (TOKAMAK, STELLARATOR, OR ALTERNATIVE) AND WHAT ADVANTAGE DOES IT PROVIDE?

2. HOW DOES YOUR DESIGN ADDRESS THE NEUTRON DAMAGE PROBLEM OVER A 30-YEAR OPERATIONAL LIFETIME?

3. WHAT Q FACTOR DOES YOUR DESIGN TARGET AND WHAT EVIDENCE FROM YOUR MODEL SUPPORTS ITS ACHIEVABILITY?

4. WHAT IS THE ESTIMATED COST AND TIMELINE FOR YOUR CONCEPT TO REACH A NET-ENERGY-PRODUCING PROTOTYPE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Plasma Physicists study superheated ionized gases and design confinement systems at national laboratories like PPPL, LLNL, and ORNL, earning $95,000-$180,000/year. Fusion Engineers work at private companies like Commonwealth Fusion Systems, TAE Technologies, Helion Energy, and General Fusion, designing and testing reactor components, earning $100,000-$175,000/year. Nuclear Materials Scientists develop radiation-resistant materials for fusion reactor walls, earning $85,000-$160,000/year.